The displacement from living cosmos to dead mechanism did not just reshape institutions and research programs. It reshaped something deeper: the cognitive infrastructure through which English-speaking cultures think. The metaphors, the categories, even the grammar of the language now carry the mechanistic framework as a default setting — embedded so thoroughly that they operate below the threshold of awareness. The framework does not just shape arguments. It shapes the mental habits that make certain arguments feel self-evident before they have been examined.

I'm not arguing that English prevents us from thinking beyond mechanism. It plainly does not — I'm writing this argument in English. My claim is more modest and more troubling: modern English makes mechanism feel like common sense. Its preferred metaphors, its explanatory habits, its grammatical defaults are comfortable with objects, parts, causes, and systems. They are less comfortable with relation, interiority, participation, and value as primary features of reality. The difficulty is not proof that such ideas are confused. It may be evidence that we are trying to think them in a language trained by the very worldview they challenge.

Metaphors as Cognitive Infrastructure

How did a methodology become a metaphysical claim about what's real? Through something deeper than explicit argument: the metaphors through which we think.

George Lakoff and Mark Johnson argued that abstract thought often operates through metaphorical mappings from concrete, embodied experience.[1] Expressions like "defending a position" or "attacking weak points" are not ornamental. They reveal a deeper metaphorical structure through which arguments are organized — not just described in terms of combat, but conducted in terms of combat. These mappings are not conscious choices. They function as part of our cognitive infrastructure — largely invisible frameworks that shape what we can perceive, what questions we can ask, and what answers seem reasonable.[2]

[1] Lakoff, George, and Johnson, Mark. *Metaphors We Live By.* United Kingdom, University of Chicago Press, 2008.

[2] Jeremy Lent has developed this insight at civilizational scale in *The Patterning Instinct* (New York: Scribner, 2017) and *The Web of Meaning* (Gabriola Island: New Society Publishers, 2021), arguing that root metaphors shape everything from social organization to ecological relationship to conceptions of flourishing.

The metaphor of the universe as a machine doesn't just change our vocabulary. It shapes what we can notice. Machines have parts, but no intrinsic purposes: whatever purposes they serve belong to their designers and users, not to the machines themselves. They can be disassembled and optimized but not participated in. They operate according to deterministic laws but lack interiority, meaning, or value. The metaphor makes certain inquiries natural — *How does it work? What are its mechanisms? What are the component parts?* — while rendering others nonsensical: *What does it mean? What is its* *purpose? What is it like to be this?*

Lakoff and Johnson help explain how metaphors structure ordinary thought. The cosmologist Edward Harrison extends the point to civilizations. Every civilization, he argued, constructs what he called a *universe* — a comprehensive model of reality so encompassing that those inside it experience it not as a model but as the world itself. These constructed universes are not the *Universe*. They are masks, fitted to the face of an unknown and perhaps unknowable underlying reality. Each mask both illuminates and obscures — enabling certain recognitions, foreclosing others, and carrying assumptions so fundamental that they become invisible to those who hold them. On Harrison's account, masks are replaced not because they fail entirely but because their exclusions eventually become too costly to ignore.[3]

[3] Harrison, Edward Robert. (1985) *Masks of the Universe.* New York, NY: Macmillan Publishing Company.

A metaphor of nature as dead matter generates a profoundly different civilizational trajectory than one of nature as a living web in which humans are embedded participants. The metaphor shapes whether Tahlequah and her pod appear to us as participants in a living reality or as complex biological machines. And that shaping largely determines what sacrifices we are willing to make to prevent their extinction.

The Framework Sets Its Own Rules

The machine metaphor's most insidious effect is not that it provides wrong answers but that it determines which questions count as legitimate.

Consider the vocabulary biology inherits for organismal behavior. When ethology describes a complex behavioral sequence — a bird's courtship display, a spider's web construction, a wolf's hunting coordination — the standard term is "fixed action pattern": a pre-programmed routine triggered by the appropriate stimulus. The term is not wrong where it fits; some animal activity really is stereotyped, and naming it precisely is useful work. The problem begins when the convention quietly trains us to treat all complex activity as pre-programmed, even in cases where the organism may be doing something closer to skilled, context-sensitive action. When an organism solves a novel problem or coordinates with others to achieve a goal none could achieve alone, the standard terms — *behavior, instinct, adaptation* — describe what happens without acknowledging that anyone is doing it. The word "response" implies passivity. The word "action" implies agency. Much biological vocabulary still defaults toward "response," especially where nonhuman organisms are concerned. The terminological choice predetermines what interpretations seem reasonable before the evidence is examined.

The pattern runs deeper than terminology for individual behaviors. It appears in the vocabulary that gene-centered biology uses to describe living systems as a whole. The terms are familiar: organisms are "genetic programs" being "executed"; behavior is "hard-wired" or "soft-wired"; brains do "information processing" through "neural circuitry"; selection produces "fitness-maximizing responses" in "survival machines." This is not a neutral technical vocabulary. Every one of these terms imports, at the level of the noun or the verb, the assumption that the phenomenon being described is fundamentally a machine running a program. An organism that appears to act with purpose is really executing code; behavior that looks like grief is really a fitness-maximizing response; interiority that seems rich and deep is really a user interface — a simplified readout of processes that are themselves purposeless. The machine metaphor enters at the level of the word and compounds from there.

This is where the machine metaphor does its deepest and least visible work. The demand for mechanistic explanation — *show the mechanism, identify the gene, specify the neural pathway* — sounds like neutral scientific rigor. But it carries a buried premise: that the only real explanations are mechanistic ones, that purpose and agency must be reduced to mechanism or conceded as illusion. I'm not arguing that mechanistic explanation is illegitimate. It is often extraordinarily powerful, and the discoveries it has produced are among our great intellectual achievements. The problem arises when a method suited to certain questions becomes the standard by which all questions are judged. The standard was constructed within the framework it appears to validate. The framework sets the rules for what counts as a good explanation, then wins by those rules. The circularity is invisible to those inside the framework, which is precisely what makes it so powerful and so difficult to challenge.[4]

[4] The argument that demands for mechanistic explanation presuppose the framework they purport to validate has several philosophical sources. Walsh, *Organisms, Agency, and Evolution*, argues that the statisticalist framework rendered organismal agency invisible by defining evolutionary explanation in terms that excluded it. Thomas Kuhn's arguments about paradigm-internal justification in *The Structure of Scientific Revolutions* (1962) remain relevant. Thomas Nagel, *Mind and Cosmos* (Oxford, 2012), extends this point specifically to consciousness. Neuroscientist Kevin Mitchell makes the point from within biology: reductionism implies that "biology is not really a science unto itself but is just complicated chemistry, and chemistry is just complicated physics," a view he calls "not just wrong — it's wrong-headed." See Mitchell, *Free Agents* (Princeton, 2023), pp. x-xi.

The Framework Occupies the Grammar

The difficulty runs deeper than metaphor and methodology. It reaches into the grammar itself.

English did not acquire its subject-verb-object structure from the mechanistic worldview. That structure is far older. But the mechanistic worldview found in SVO grammar an unusually comfortable home: discrete things performing actions upon other discrete things. The grammar makes substance-based metaphysics feel like a direct description of how reality is structured, and process-based metaphysics feel like an unnatural contortion of that description. When I try to discuss consciousness, I find I must cast it as either a thing that exists, a process that happens to someone, or a property that something has. English can express the alternative — consciousness as dynamic relation rather than as a thing inhabited — but usually only by strain, abstraction, or workaround. The evidence will point, in later chapters, toward something more like what process philosophy describes: reality constituted by dynamic relations rather than static substances. The grammar is not impossible to escape, but it's an uphill climb.

Consider Tahlequah. To describe what she did during those seventeen days, English grammar requires casting her as a subject (an orca) performing actions (carrying, diving, pushing) on an object (her dead calf). The grammar handles the observable sequence. But it doesn't naturally hold the reality of her interior state — her sustained bereavement, her refusal to let the calf go, the shape of her undivided grief. It catches the external form but struggles to register, without elaborate compensation, that what we observed is only a surface expression of the real phenomenon.

The depth of the linguistic reshaping can be traced in the history of a single word — or rather, in the history of a word that was split in two.

"Consciousness" and "conscience" were historically the same word. In Latin, *conscientia* meant "knowing together" — *con-scire*, a knowing held in common with others. To be conscious was to be accountable to what one knows. When medieval philosophers used *conscientia*, they did not distinguish between awareness and moral sensitivity because the distinction seemed artificial. To know something was already to stand in evaluative relationship to it.

French preserved this unity. *Conscience* still covers both meanings — awareness and moral sense — requiring context to distinguish them.

English gradually separated what Latin *conscientia* and French *conscience* could hold together: awareness and moral self-relation. By the late seventeenth century, "consciousness" was emerging as a more technical term for reflexive awareness and personal identity, while "conscience" was increasingly reserved for the moral sense. The separation was not a single event. It unfolded across decades, driven by theological, philosophical, and psychological currents running partly alongside and partly against the mechanistic turn. But the mechanistic turn gave the separation a particular shape. It supplied a reason to want a term for mind as an object of study — one stripped of the evaluative weight that *conscientia* had carried when mind was understood as a moral witness rather than a mechanism.[5]

Locke's *Essay Concerning Human Understanding* (1690) gave philosophical force to this direction of travel, using "consciousness" as a technical term for the reflexive awareness that constitutes personal identity across time, bracketing much of the older moral-theological

[5] For the English semantic shift specifically, see C.S. Lewis, *Studies in Words*, 2nd ed. (Cambridge: Cambridge University Press, 1967), chapter 8 ("Conscience and Conscious"). Lewis traces the gradual division across the seventeenth century, before and around Locke, and notes that the older sense of "conscious" — knowing together with another, or with oneself as moral witness — was still active in seventeenth-century usage even as "consciousness" was emerging as a more technical term for reflexive awareness.

resonance the word had carried.[6] German made the distinction unusually explicit: Christian Wolff coined *Bewusstsein* (consciousness) as a technical term deliberately distinct from *Gewissen* (conscience), though both share the root *wissen*, to know.[7]

The implications extend beyond etymology. If the pre-mechanistic word was right — if awareness is inherently evaluative, if to be conscious is already to stand in relationship to what one knows — then the mechanistic framework did not simply provide a more precise vocabulary. It provided a narrower one, making it difficult to express in a single word what the earlier term had held together: awareness as inherently evaluative.

This is not merely an etymological curiosity. It points toward one of this essay's central claims: consciousness is not bare registration. To be aware is already to be situated, responsible, and in relation.[8] What the earlier word named was not awareness plus evaluation, but awareness-as-evaluation — a single capacity the mechanistic turn learned to divide into a descriptive faculty (consciousness) and a separate normative one (conscience).

When I turn, in a later chapter, to what Thomas Metzinger calls a *Bewusstseinskultur* — a culture of consciousness — I'll be pointing toward something closer to the original *conscientia* than to the modern "consciousness" the mechanistic turn produced: awareness that knows itself, cares about what it knows, and takes responsibility for that knowing. Not consciousness or conscience as separate faculties, but consciousness as conscience — awareness that is constitutively, not contingently, evaluative. The Latin word may have been more adequate to the reality than the English word that replaced it.

Other Languages, Other Resources

Languages and knowledge traditions less shaped by early modern European mechanism often preserve conceptual resources that English handles awkwardly. In Chinese, the character — *xīn* — functions as both "heart" and "mind," refusing the separation between emotion and cognition that English makes nearly automatic. The point is not that Chinese thought lacks distinctions — it has many, including technical philosophical vocabularies for separating cognition, emotion, and intention — but that the basic word doesn't divide what English forces apart.

Two North American Indigenous languages illustrate distinct grammatical resources English handles awkwardly. The grammar of animacy in Potawatomi, which Robin Wall Kimmerer has described, makes it harder to collapse living beings and inert objects into the same grammatical category: the language carries forward a distinction

[6] John Locke, *An Essay Concerning Human Understanding*, ed. Peter H. Nidditch (Oxford: Clarendon Press, 1975 [1690]). Book II, Chapter XXVII ("Of Identity and Diversity") makes "consciousness" central to Locke's account of personal identity across time. Locke is not coining the term but giving it the philosophical role for which the *Essay* is famous.

[7] Metzinger, T. (2024). The Elephant and the Blind: The Experience of Pure Consciousness. Cambridge, MA: MIT Press. The epilogue on *Bewusstseinskultur* discusses the consciousness/conscience connection and its implications for mental autonomy.

[8] **?** Develops enactive framework showing consciousness as inherently value-laden and concerned.

between things that can act and reciprocate and things that cannot, a distinction that English makes only weakly through the he/she/it system: a living orca and a rock receive the same pronoun: "it." Potawatomi grammar makes this conflation structurally impossible.[9] The physicist F. David Peat, drawing on extended engagement with Indigenous thought, has described Blackfoot as more verb-centered than English, organized around processes and relationships rather than around things. The point is not that English cannot express process, but that it has to work harder to do so. English typically requires process to be nominalized — turned into a noun-form — before it can function as a sentence's subject.

[9] Robin Wall Kimmerer, *Braiding Sweetgrass: Indigenous Wisdom, Scientific Knowledge and the Teachings of Plants* (Minneapolis: Milkweed Editions, 2013). Kimmerer's discussion of the "grammar of animacy" appears especially in the chapter "Learning the Grammar of Animacy." For more technical treatments of grammatical animacy in Algonquian languages including Potawatomi, see Marianne Mithun, *The Languages of Native North America* (Cambridge: Cambridge University Press, 1999), chapter 3.

The point extends beyond individual examples. Tewa scholar Gregory Cajete has argued, in *Native Science*, that many Indigenous knowledge systems integrate empirical observation with relational participation rather than treating them as opposed. What Western mechanistic frameworks often dismiss as anthropomorphism — projecting human qualities onto nature — many Indigenous thinkers describe instead as a disciplined perception of relationship, agency, and reciprocity, developed across generations of close observation of specific places and the beings who live in them.

Rather than enlisting these traditions as evidence for an argument built elsewhere, I'm noting that traditions developed independently of the Western mechanistic turn arrived at recognitions that the post-mechanistic Western project is now slowly recovering — and that this convergence matters. Vine Deloria Jr. and contemporary philosopher Kyle Powys Whyte have shown that Indigenous philosophical traditions engage Western philosophy and science seriously, on their own terms. Their common ground with this essay's project — reality constituted by relationships rather than isolated substances, multiple valid modes of knowing, interiority as fundamental rather than derivative — is the subject of later chapters.[10]

[10] Gregory Cajete, *Native Science: Natural Laws of Interdependence* (Santa Fe: Clear Light Publishers, 2000). Vine Deloria Jr., *The World We Used to Live In: Remembering the Powers of the Medicine Men* (Golden, CO: Fulcrum Publishing, 2006). Kyle Powys Whyte's work on Indigenous environmental philosophy and cross-traditional epistemology appears in numerous venues; see especially "Indigenous Science (Fiction) for the Anthropocene: Ancestral Dystopias and Fantasies of Climate Change Crises," *Environment and Planning E: Nature and Space* 1, no. 1–2 (2018): 224–242.

The Daoist vocabulary makes a related point about the limits of grammatical reification. The *Dao De Jing*'s opening line — *the Dao that can be named is not the enduring Dao* — builds a warning into its own founding text against treating its central word as a thing the word names. This is linguistic precision about the limits of linguistic precision: a recognition that some realities resist the thing-with-properties categories Indo-European grammar takes for granted.

Heisenberg's phrase "central order" shows a similar linguistic pressure at work in twentieth-century physics: the need for a term that could point toward order without reifying it into a thing, and the difficulty of finding such a term in the grammatical inheritance available.

These are not exotic curiosities. They are reminders that conceptual possibilities are not constrained by English. Other languages and traditions preserve distinctions, fusions, and relational grammars that can make visible what modern English tends to obscure. The difficulty we keep encountering may not lie only in the ideas. It may also lie in the inherited language through which we're trying to think them.

The constraint operates even on this essay's own key terms. The Prologue introduced "interiority" to name the experiential dimension of existence — what it is like, from the inside, to be a particular organism engaging with its world. But notice what English has already done: "interiority" is a spatial noun, a container; "experiential" is an adjective, a quality of something else. The grammar forces them into separate roles — a thing and its attribute — though the phenomenon they point to is single and undivided. There is no place in an organism's felt engagement with its world where the interior stops and experience begins. The separation is an artifact of the grammar, not of the reality. This difficulty will surface again in later chapters, at moments where the argument presses hardest against the boundaries of what English can hold.

The quantum physicists faced a related difficulty a century ago, and they did not all frame it the same way. Bohr insisted that the results of quantum mechanics could only be communicated in the language of classical physics, even though the classical concepts were inadequate to the quantum domain. "We are suspended in language," he is reported to have said.[11] Pauli identified a sharper version of the problem. For him, quantum mechanics was a domain where ordinary language and pictorial intuition simply did not apply. Electron spin was the first property in the history of physics that could not be visualized at all. The mathematics carried what no picture and no ordinary description could hold. Heisenberg made a related move when he set aside the attempt to picture what electrons were doing inside atoms and worked only with what could be observed and calculated.[12]

The physicists were not the first to encounter this. Contemplative traditions have reported for millennia that certain experiences are structured, precise, and cognitively rich — yet resist description in ordinary language. Phenomenologists from Husserl through Merleau-Ponty encountered the same limit when they pressed hard enough against the description of lived experience. Poets and mystics across cultures have mapped a region where linguistic approximations are no longer useful, even as language remains the medium of whatever communication is possible at all. What the physicists added was one more confirmation of a condition that may be built into the relationship between human language and certain features of reality: language is always part of thinking, and sometimes it runs out.

This is the situation any serious engagement with experiential awareness eventually meets. One has no choice but to use language

[11] Aage Petersen, "The Philosophy of Niels Bohr," *Bulletin of the Atomic Scientists*, vol. 19, no. 7 (September 1963), p. 10.

[12] On Pauli's view of spin as a fundamentally non-visualizable property, see Wolfgang Pauli, "Exclusion Principle and Quantum Mechanics" (Nobel Lecture, 1946), in *Writings on Physics and Philosophy*, ed. Charles P. Enz and Karl von Meyenn (Berlin: Springer, 1994). For Heisenberg's account of the shift away from visualization in matrix mechanics, see Werner Heisenberg, *Physics and Philosophy: The Revolution in Modern Science* (New York: Harper, 1958), especially chapters 2 and 3. On the broader epistemological significance of the move away from visualization in early quantum theory, see Arthur I. Miller, *Imagery in Scientific Thought: Creating 20th-Century Physics* (Boston: Birkhäuser, 1984).

— abandoning it would be abandoning inquiry itself. But one also has no choice but to recognize that language is not coextensive with what can be known or experienced. The two conditions are complementary in the sense Bohr gave the term: mutually necessary, mutually incomplete. Every tradition that has worked seriously at this edge has developed its own way of holding the complementarity — the physicists with mathematics, the contemplatives with ritual and disciplined attention, the phenomenologists with their reductions, the poets with the lyric.